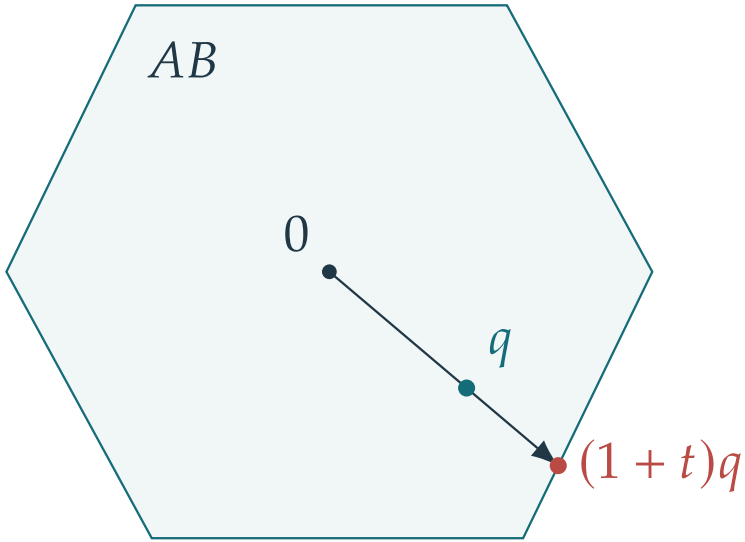
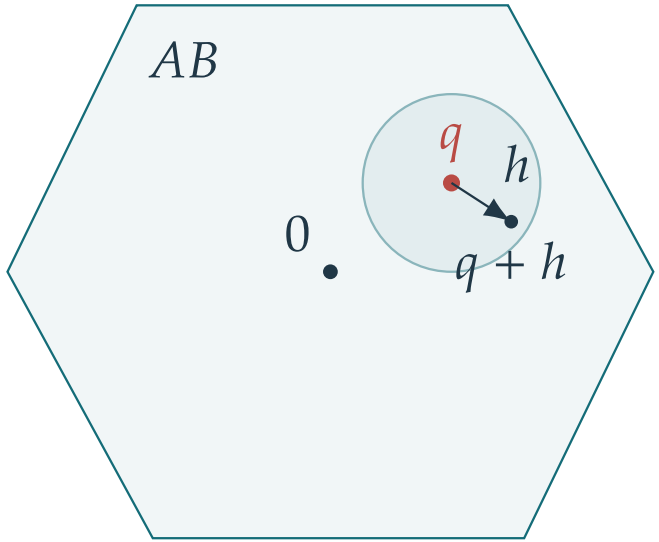
Radial extension from an interior point



$Au = (1+t)q,\ u \in B$
$z = ru/(1+t)$ gives
$Az = Ax_0$ and $\|z\|_1 < r$.

A better feasible vector gives an interior point



$Az = Ax_0,\ \|z\|_1 < r$
$q + h = A(z/r + A^+ h) \in AB$
for small h.

$r = \|x_0\|_1 > 0, \qquad q = Ax_0/r, \qquad A$ onto